



Selenium & Playwright Course Content (Total Duration : 24 days)

Required Software :

Jdk 17
Excel
Eclipse for Java Developers – V 2024-12
GitBash
Chrome, Edge
Maven
Claude AI - local

Core Java, HTML & CSS – 2.5 days

Core Java :

Introduction to all predefined packages
More about JVM, JRE & JDK
Types of variables
Data Types in Java
Introduction to OOPS
 Class
 Object
 Local, Instance and static variables
 Constructors
 Methods
 Arrays
 This & super keyword
 Final Keyword
 Inheritance and Types
 Working with super classes
 Using types of polymorphism such as overloading, overriding, and dynamic binding
 Abstraction
 Encapsulation
 Abstract Classes
 Interfaces
String Methods
Collections – ArrayList, Set, Map
Exception Handling
Reading from external file – Property file
HTML & CSS

Selenium : - 5.5 days

Introduction to Automation
Introduction to Selenium
 Selenium Overview
 Features of Selenium

Maven Overview
 Introduction To Maven Goals
 Creating a Maven Project
 Setting up Maven project with Selenium Dependency

Selenium WebDriver 4 API



Introduction to Selenium WebDriver
WebDriver Architecture

Interacting with WebElements

Button, TextBox, TextArea, Dropdown, RadioButton, Checkbox, ToolTip

Locators

Locator Strategies - Id, name, linkText, PartialLink, tagName, cssselector, xpath, classname

Relative Locators

Dynamic xpath

Handling Alerts

Handling Multiple Windows

Handling Web Tables

Handling iFrames

Interacting with ShadowDOM

Synchronization

Using Javascript Executor

TestNG Framework

Introduction to TestNG framework

TestNG Annotations

Creating and configuring Testng.xml file

TestNG reports

Running failed testcases

Running TestNG test as Maven test

TestNG Options

Priority

Groups

Parameterization

Implementing DataDriven Framework using .CSV file as Dataprovider

Reading from External Files

Using property file

Using excel as Object Repository

Extent Reports

Generating Extent Reports

Attaching screenshots with ExtentReports

Selenium Grid

Components in Selenium Grid

Running in Standalone and Hub&Node mode

Implementing Design Pattern – Page Object Model

Page Object Model

Page Factory

Implementation of POM & PF

Framework Design

Automating any eCommerce portal using POM by implementing a framework.

API Testing using RestAssured – 1 day

Rest API Testing - Get, POST, PUT, DELETE

Post using .json payloads



API request using Authentication

BDD & Cucumber – 1.5 days

- Introduction to BDD & Cucumber
- Feature Files, Step Definition & Runner (TestNG)
- Default Reports, Rerunning failed testcases
- Scenario Outline, Table, Parallel Execution
- Hooks, Tags, Background

Version Control – .5 day

- Working with Git and GitHub
- Basic Git commands - init, add, commit, push, pull, clone
- CI/CD Integration
 - Introduction to CI/CD pipeline
 - Running Selenium test from Jenkins and generating reports
- Casestudy Evaluation – 2 days

Playwright with Typescript (Duration : 11 days)

Prerequisites :

Web application manual testing knowledge is a must and knowledge in any programming language

Required Software :

- NodeJS V22+
- Typescript 5.2.2, Java 17
- Npm
- VS Code V1.101
- GitBash
- Chrome, Edge
- Claude AI

Detailed Course Contents :

TypeScript Introduction & Environment Setup

- Javascript - Callback functions, Promises, Async / Await
- What is TypeScript?
- Setting up Node.js & VS Code

Typescript – 2 days

- Transpiling .ts to .js
- Data types - string, number, boolean, any, void, union

Functions

- Function declarations & expressions
- Optional and default parameters
- Arrow functions
- Callback functions

Arrays & Tuples

Class Features

- Creating classes



Readonly properties

Inheritance & Access Control

extends, method overriding

Access modifiers: public, private, protected

Importing/exporting modules

Playwright – 5.5 days

Introduction to Playwright

Installing Playwright

Initializing Playwright project

Understanding project structure

Writing first test

Running tests (headless/headed mode)

Running specific tests & multiple tests

UI Mode overview

Playwright Locators

[getByText, getByRole, getByLabel](#)

[locator\(\), page.locator\(\)](#)

Basic Actions

[Typing text, clicking buttons](#)

[Handling radio buttons, checkboxes](#)

[Dropdown](#)

[Tooltips](#)

iFrames

Locating elements inside iframes

Capture Text of the Elements

[innerText\(\) vs.textContent\(\)](#)

[allInnerTexts\(\) vs.allTextContents\(\)](#)

[all\(\) Method](#)

Dialogs, Windows & Tabs

Handling alerts, prompts, confirm dialogs

Switching between tabs/windows

Mouse Actions

Hover, right-click, double-click, drag & drop

Scrolling

Scrolling to elements

Infinite scrolling techniques

File Upload & download

Browser Context & Pages

Working with new contexts

Multi-tab and isolated sessions

Auto-waiting, Timeouts & Assertions

Auto-waiting mechanism

Test and Expect Timeouts

Assertions (Auto-retrying & Non-retrying assertions)

Record the tests (Test Generator/Codegen)

Record the tests using Codegen

Identify locators in Codegen

Modifying generated tests

**Debugging tests**

- Live debugging, editing locators
- Run in Debug Mode
- Picking a Locator

Screenshots & Videos

- Capturing screenshots
- Recording test runs
- Viewport

Tracing with Trace Viewer

- Enabling tracing
- Opening trace files for debugging
- Retries and handle Flaky tests

Grouping Tests & Hooks

- grouping tests with describe()
- Hooks - beforeAll, afterAll, beforeEach, afterEach
- Annotations
- Tagging Tests

Parallel Test Execution

- Setting parallelism in config
- Worker-level isolation & Limiting workers

Parameterization

- Different ways to Parameterize data to the tests
- Data Driven Testing with External files(JSON)

Reporting

- Default reports

Page Object Model (POM)

- Creating reusable page classes
- Encapsulating page actions

Visual Testing/Visual Comparisons

- Generating Screen shots
- Comparing snapshots

API Testing with Playwright - 1 Day

- Rest API Testing - Get, POST, PUT, DELETE
- Post using .json payloads
- API request using Authentication

CI/CD Integration

- Introduction to CI/CD pipeline
- Running Playwright test from Jenkins and generating reports

Casestudy Evaluation – 2 days**AI Driven - .5 day**

- Configure Selenium & Playwright MCP Server in Claude AI and generate testscripts.